Training a small image model

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Training a small image model before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Pooling and spatial information

Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Image tensors and channels

Without opening the old lesson, explain one key idea from Image tensors and channels. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Initialisation and reproducibility

Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Use fictional 2 by 2 binary images, stored as two rows. Predict class 1 when the left column sum exceeds the right column sum; otherwise predict class 0. Equal sums belong to class 0.

This known arithmetic task makes every step checkable; it does not establish real-world image recognition.

Split and ID	Image rows	Class	Difference d
Train A	[1, 0] and [0, 0]	1	1
Train B	[1, 0] and [1, 0]	1	2
Train C	[0, 1] and [0, 0]	0	-1

Split and ID	Image rows	Class	Difference d
Train D	[0, 1] and [0, 1]	0	-2
Validation E	[0, 0] and [1, 0]	1	1
Validation F	[0, 0] and [0, 1]	0	-1

Apply the fixed kernel [[1, -1], [1, -1]] to the entire image to obtain d. Its negative gives -d. ReLU produces h1 = max(0, d) and h2 = max(0, -d). The **output** score is a(h1 - h2) = ad.

Only the shared scalar a is trained; both kernels stay frozen and there is no bias or pooling.

For squared-error **training**, map class 1 to target t = 1 and class 0 to t = -1. Predict class 1 only when score > 0; otherwise class 0. The score is not a probability.

Keep the teacher-held final cases unseen until the **checkpoint** is selected and recorded.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Training a small image model trace and record the first step you can explain without help.

Use W2: Check both updates for **Training** a small image model. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Training a small image model explanation against the supplied facts before discussing it.

Use the worked case for **Training** a small image model. Proposed explanation: “All parameters in a trained model must have been learned”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Choose a checkpoint independently

A separate model scores bd. Its **training** examples have d = 1 and -1 with signed targets 1 and -1. Start b = 0; gradient is 2(b - 1), rate 0.25 and budget two updates. Validation differences are 2 and -2 with targets 1 and -1.

Calculate all three weights and their validation MSEs, then select by lowest validation MSE.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Trace features and a prediction

> Trace features and a prediction. Apply the supplied Training a small image model facts and keep your reasoning.

For Train B, compute d, h1, h2 and the score at a = 0.45. Give the class **prediction** and squared error against its signed target.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Check both updates

> Check both updates. Apply the supplied Training a small image model facts and keep your reasoning.

Starting at a = 0, use gradient 5a - 3 and rate 0.1 for two updates. Give each gradient, new weight and **training** MSE.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Apply the selection rule

> Apply the selection rule. Apply the supplied Training a small image model facts and keep your reasoning.

Select among the three saved lesson **checkpoint**s using validation MSE, with earliest update breaking a tie. Is selecting by **training** accuracy the same declared procedure?

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Choose a checkpoint independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Choose a checkpoint independently. Apply the supplied Training a small image model facts and keep your reasoning.

A separate model scores bd. Its **training** examples have d = 1 and -1 with signed targets 1 and -1. Start b = 0; gradient is 2(b - 1), rate 0.25 and budget two updates. Validation differences are 2 and -2 with targets 1 and -1.

Calculate all three weights and their validation MSEs, then select by lowest validation MSE.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Audit a final evaluation

> Audit a final evaluation. Apply the supplied Training a small image model facts and keep your

> reasoning.

After inspecting the final cases, a team changes the decision threshold and reports the best result as an unseen test score. Explain the problem and list two things the original report should preserve.

> **My reasoning and evidence**

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Training a small image model to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled **training** a small image model **evidence** sheet to Image-pattern Network Investigation. Show how you can trace a complete image model with fixed features and one trained weight.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Training a small image model

> Train one tiny image score with a fixed budget and a separately reserved check. Explain the labelled step before checking the changed condition.

Train one tiny image score with a fixed budget and a separately reserved check.

Toy brightness difference d=left–right; score=a×d. **Training** pair d=2, target 2; start a=0; squared loss; learning rate 1/16; two updates.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/23

Training a small image model

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. First update

Gradient 2(2a-2) x 2=-8; new a = 0.5, training loss 1.

2. Second update

Gradient at 0.5 is-4; new a = 0.75, training loss 0.25.

3. Validation score

At reserved d1, target 1: prediction 0.75, squared error 0.0625.

CHANGE ONE THING

Double the two-update budget while keeping the same training rule.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Does lower **training** loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week’s input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Train one tiny image score with a fixed budget and a separately reserved check.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? At reserved d1, target 1: **prediction** 0.75, squared error 0.0625.

1. Validation score

2. First update

3. Second update

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Validation score At reserved d1, target 1: **prediction** 0.75, squared error 0.0625.

The image feature is a brightness difference, and **training** adjusts one **output** weight multiplying it. Each gradient is computed using the current **prediction** before the next update.

In this example, the rate halves the distance to the **training** optimum each time. More updates reduce this **training** loss, but choosing a budget still requires separate validation **evidence**.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Double the two-update budget while keeping the same **training** rule. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Next a values 0.875 and 0.9375; **training** loss falls to 0.015625.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625. Each update halves the distance to the **training** optimum 1 in this toy system.

Repair a missing building block

Review Pooling and spatial information (ai-5/22). Goal: Compute max and mean pooling under an explicit window rule. Short practice: Can the max map reveal where the 4 originally appeared?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Image tensors and channels (ai-5/20). Goal: Read a pixel using declared row column and channel axes. Short practice: Where is the bottom green pixel in the first batched image?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Initialisation and reproducibility (ai-5/17). Goal: Trace how starting values affect a fixed **training** budget. Short practice: Does the matching final loss mean the parameter values are equal?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

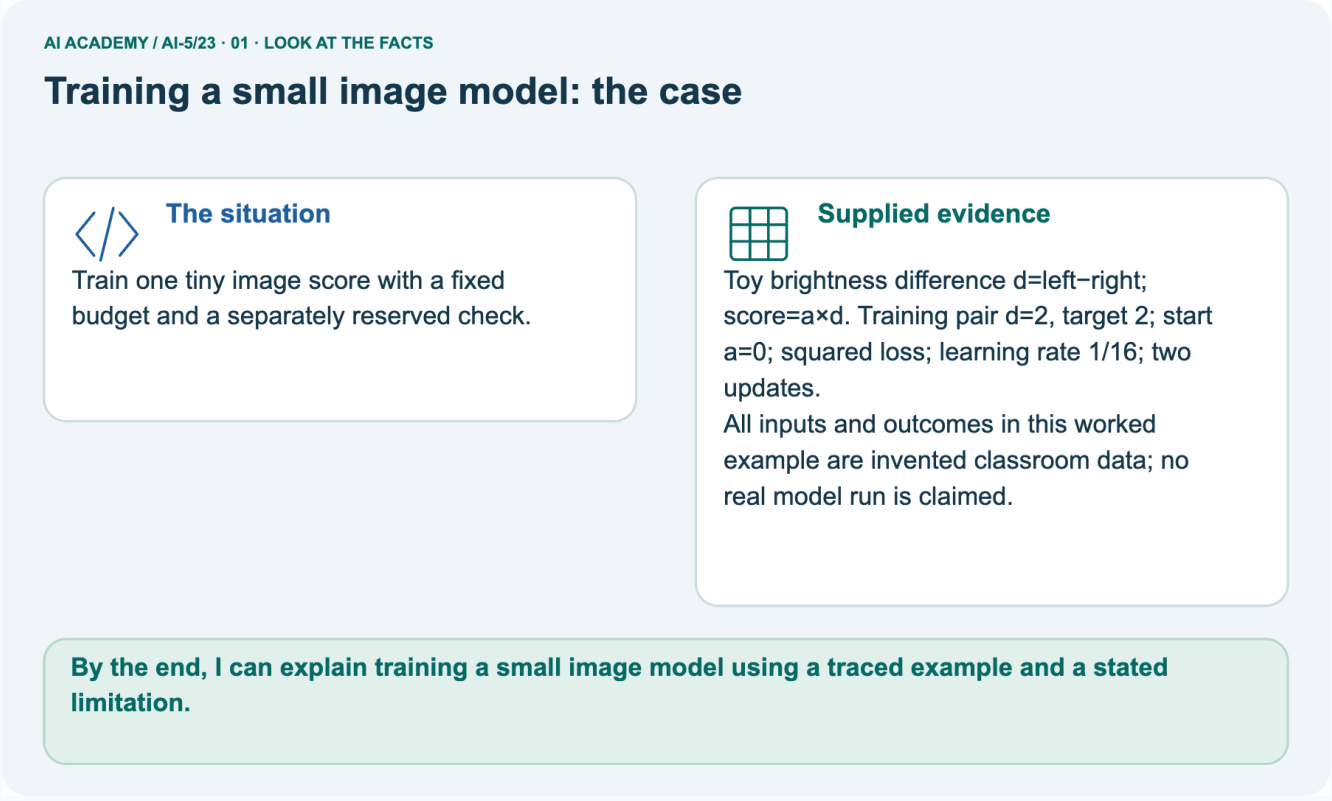
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Training a small image model picture from its labels, then record a relevant limit.

01. Training a small image model: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

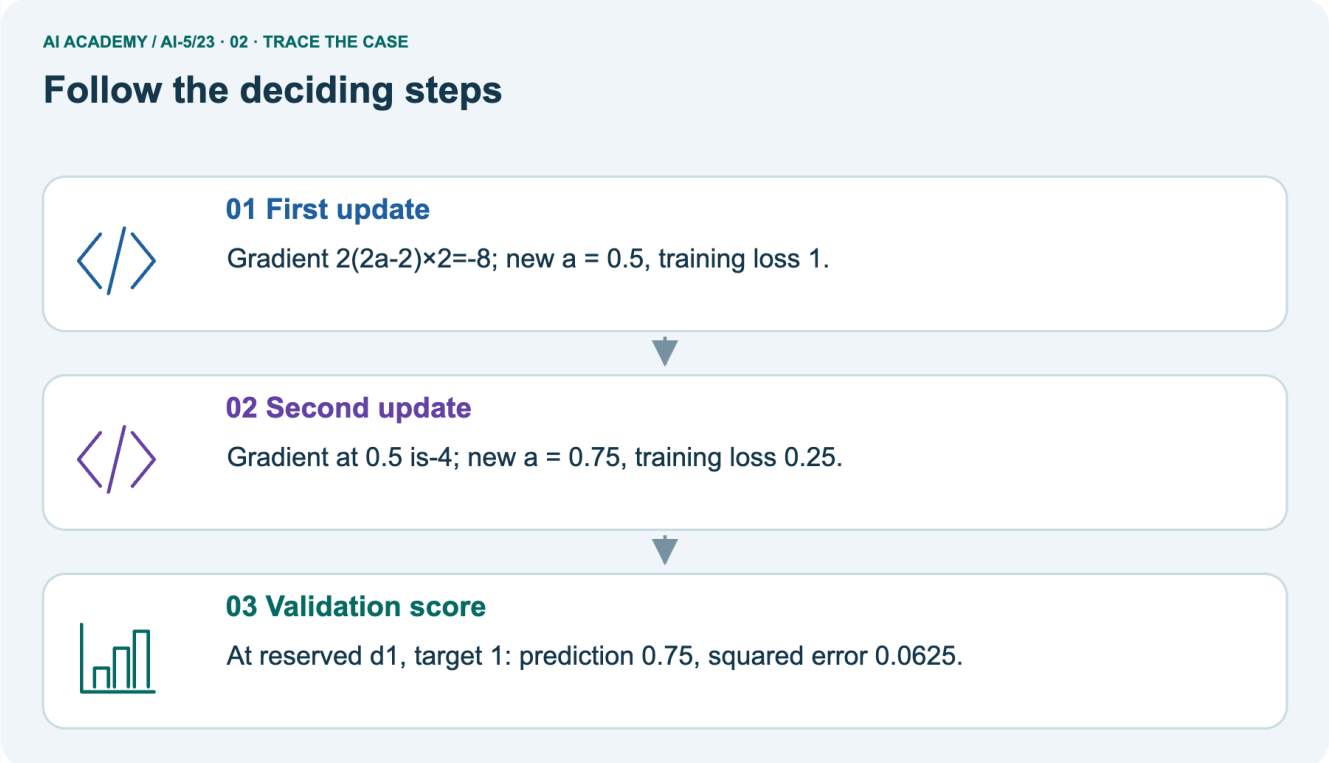
Train one tiny image score with a fixed budget and a separately reserved check. Supplied facts: Toy brightness difference d=left−right; score=a×d.

Training pair d=2, target 2; start a=0; squared loss; learning rate 1/16; two updates. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



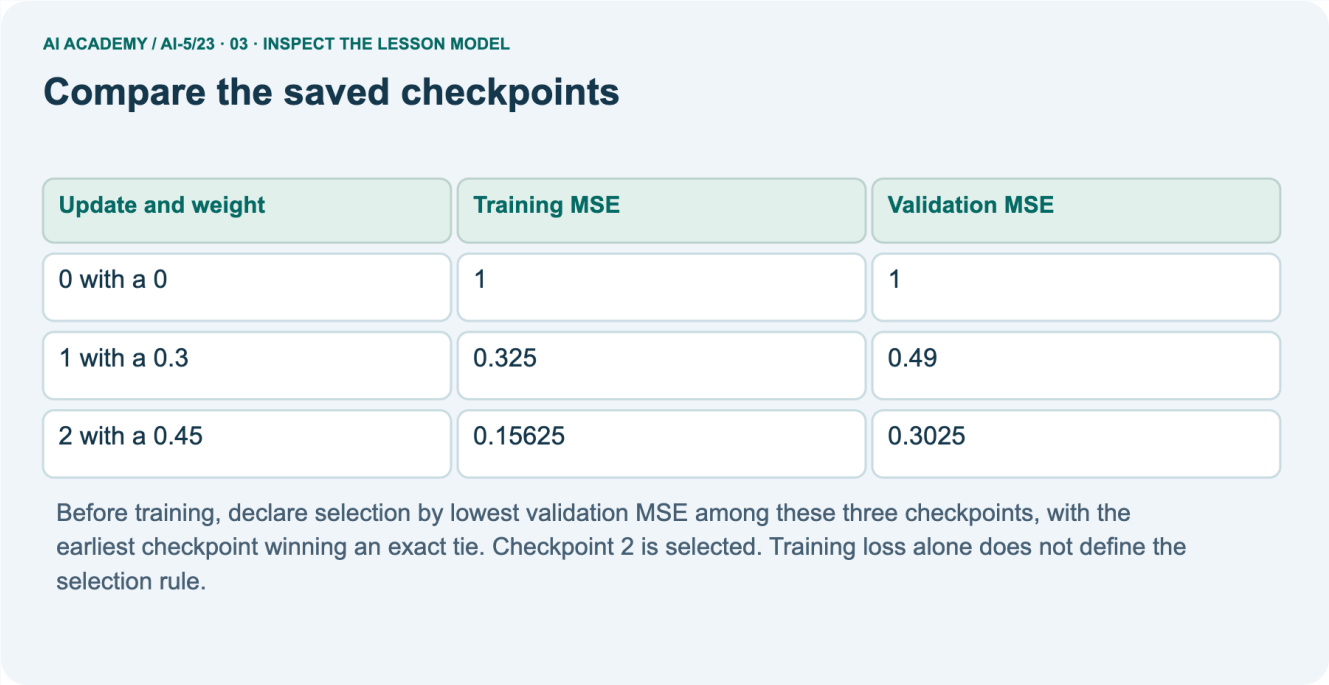
Follow the labelled picture with the matching explanation. Open the vector diagram

First update: Gradient 2(2a-2)×2=-8; new a = 0.5, **training** loss 1. → Second update: Gradient at 0.5 is-4; new a = 0.75, **training** loss 0.25. → Validation score: At reserved d1, target 1: **prediction** 0.75, squared error 0.0625.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Compare the saved checkpoints



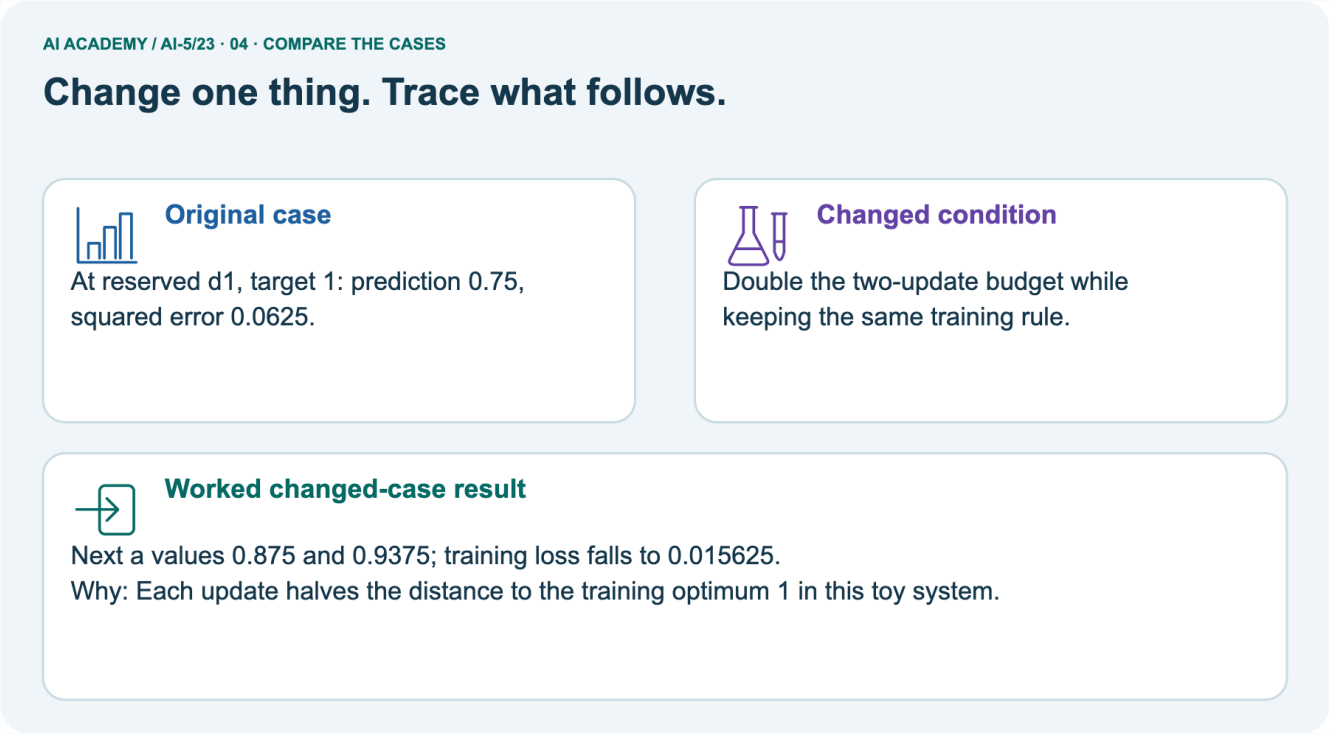
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare the saved **checkpoint**s. 0 with a 0 | 1 | 1. 1 with a 0.3 | 0.325 | 0.49. 2 with a 0.45 | 0.15625 | 0.3025

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

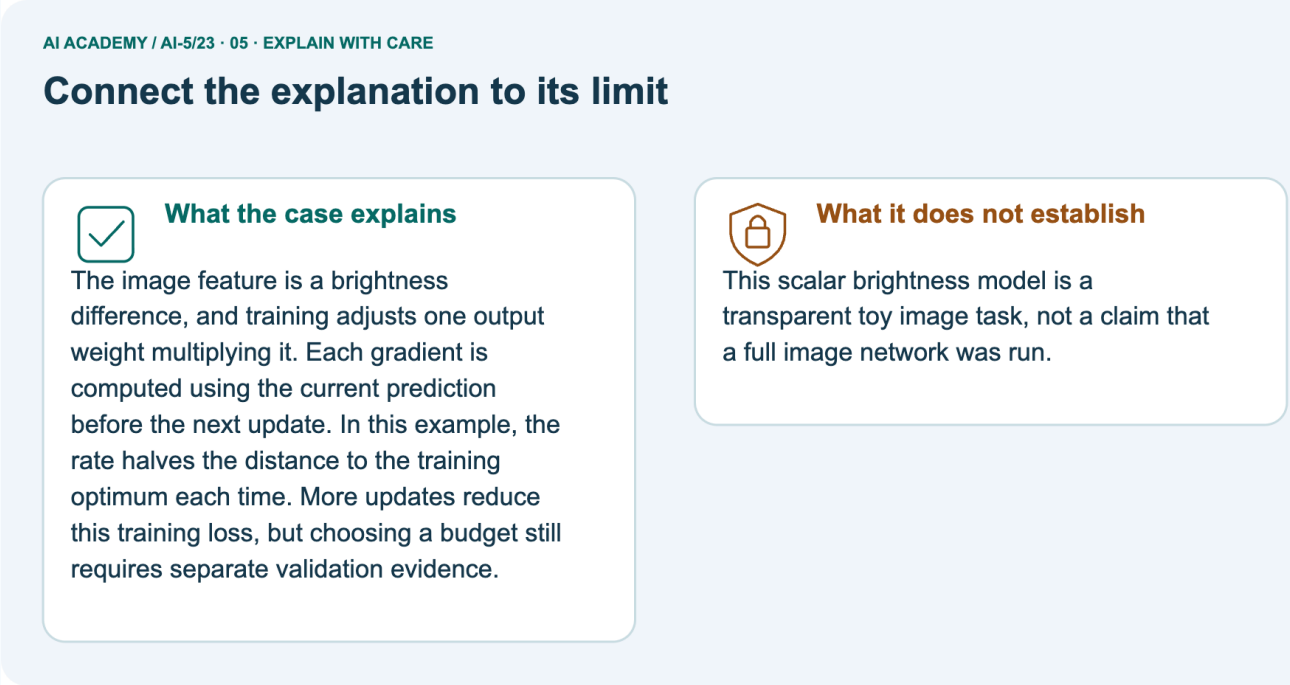
Original case: At reserved d1, target 1: **prediction** 0.75, squared error 0.0625. Changed condition: Double the two-update budget while keeping the same **training** rule.

Expected result: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625. Reason: Each update halves the distance to the **training** optimum 1 in this toy system.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The image feature is a brightness difference, and **training** adjusts one **output** weight multiplying it. Each gradient is computed using the current **prediction** before the next update.

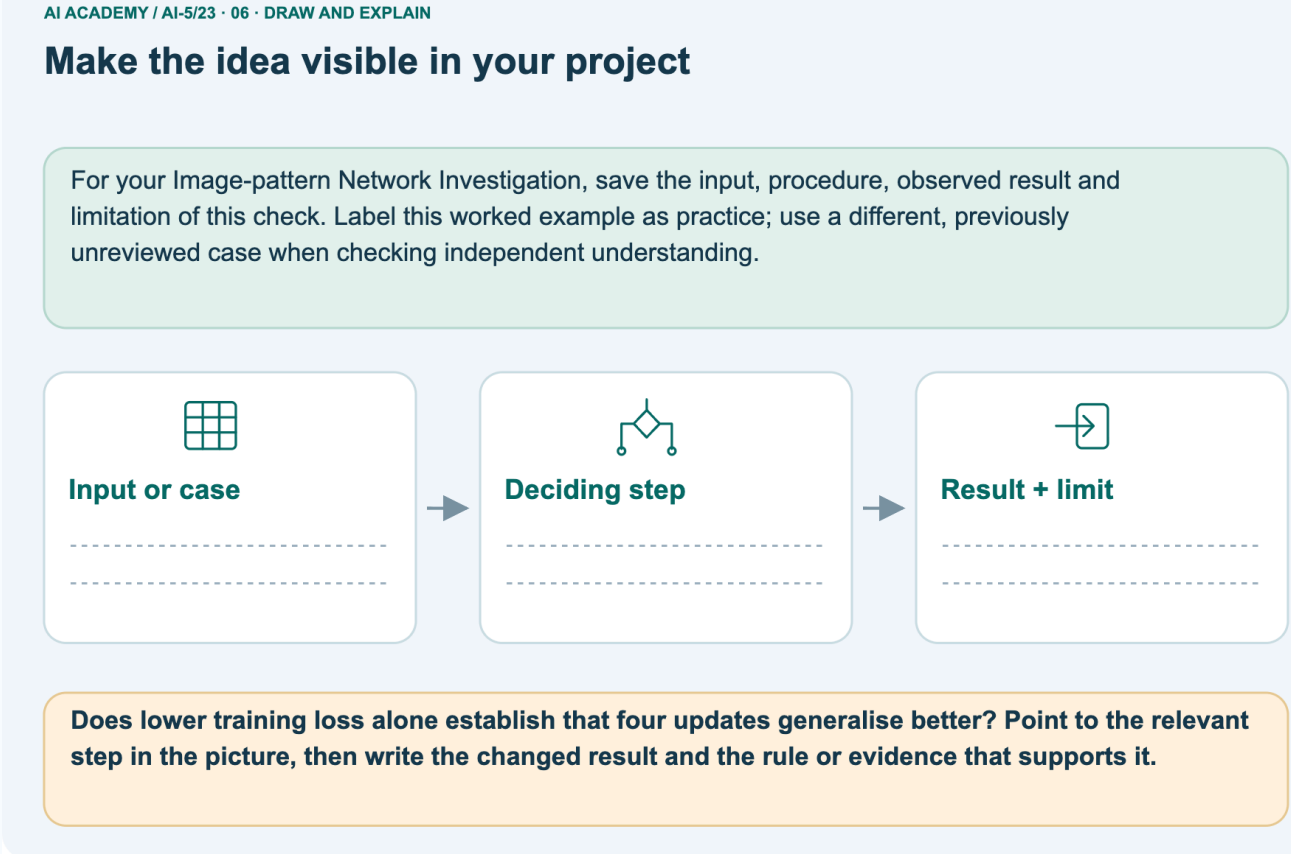
In this example, the rate halves the distance to the **training** optimum each time. More updates reduce this **training** loss, but choosing a budget still requires separate validation **evidence**.

Limit: This scalar brightness model is a transparent toy image task, not a claim that a full image network was run.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does lower **training** loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.